

CENTRAL BOARD OF SECONDARY EDUCATION**HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)****SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 7: WORK, ENERGY, AND SIMPLE MACHINES****TIME ALLOWED: 45 MINUTES****MAXIMUM MARKS: 25****PAPER CODE: 086/CH07****Student's Name:** _____**Roll No.:** _____**Class & Section:** Class IX - _____**Date of Exam:** _____**Invigilator's Sign:** _____**GENERAL INSTRUCTIONS:**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** Standard Multiple Choice Questions with one correct option.
3. **Section B (Q16 – Q20):** Assertion-Reasoning based questions. Read both statements carefully before marking.
4. **Section C (Q21 – Q25):** Case-Study / Practical Data based questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** using a blue or black ballpoint pen.

SECTION A: MULTIPLE CHOICE QUESTIONS (Q.1 TO Q.15)**[15 MARKS]**

Q.1 A porter lifts a luggage of 15 kg from the ground and puts it on his head 1.5 m above the ground. Taking $g = 10 \text{ m/s}^2$, the work done by him against gravity is:

- (A) 150 J (B) 225 J
(C) 22.5 J (D) 15 J

Q.2 A satellite revolves around the Earth in a circular orbit under the gravitational pull of the Earth. What is the work done by gravity on the satellite in one complete revolution?

- (A) Positive and equal to kinetic energy (B) Zero
(C) Negative (D) Infinite

Q.3 What happens to the kinetic energy of a moving object if its speed is doubled?

- (A) It is doubled (B) It becomes four times its initial value
(C) It is halved (D) It remains unchanged

Q.4 The work done by friction on an object sliding across a rough floor is always:

- (A) Positive (B) Negative
(C) Zero (D) Imaginary

Q.5 An electric bulb consumes 1000 J of electrical energy in 10 seconds. The power of the bulb is:

- (A) 100 W (B) 10,000 W
(C) 10 W (D) 0.1 W

Q.6 One commercial unit of electrical energy (1 kilowatt-hour or 1 kWh) is equal to how many Joules?

- (A) $3.6 \times 10^5 \text{ J}$ (B) $3.6 \times 10^6 \text{ J}$
(C) $3.6 \times 10^4 \text{ J}$ (D) $1.0 \times 10^3 \text{ J}$

Q.7 A body of mass 2 kg is dropped freely from a height of 20 m. Its kinetic energy just before striking the ground is ($g = 10 \text{ m/s}^2$):

- (A) 200 J (B) 400 J
(C) 40 J (D) 800 J

Q.8 The mechanical advantage (MA) of a simple machine is defined as the ratio of:

- (A) Effort to Load (B) Load to Effort
(C) Work output to Work input (D) Distance moved by effort to distance moved by load

Q.9 In a Class 1 lever, which component is located in the middle between Load and Effort?

- (A) Fulcrum (B) Load
(C) Effort (D) Resistance

Q.10 A wheelbarrow and a nutcracker are classic examples of which class of lever?

- (A) Class 1 lever (B) Class 2 lever
(C) Class 3 lever (D) Compound pulley

Q.11 An electric motor takes 2 minutes to lift a 200 kg mass through a vertical height of 30 m. The power of the motor is ($g = 10 \text{ m/s}^2$):

- (A) 500 W (B) 60,000 W
(C) 250 W (D) 1000 W

Q.12 A compressed spring possesses which form of potential energy?

- (A) Gravitational potential energy (B) Elastic potential energy
(C) Chemical energy (D) Thermal energy

Q.13 A force of 10 N displaces an object by 5 m at an angle of 60° to the direction of the force. The work done is ($\cos 60^\circ = 0.5$):

- (A) 50 J (B) 25 J
(C) 0 J (D) 100 J

Q.14 Which of the following devices transforms chemical energy directly into electrical energy?

- (A) Electric generator (B) Dry cell / Electric battery
(C) Solar cell (D) Electric toaster

Q.15 An ideal single fixed pulley has a mechanical advantage of:

- (A) 1 (B) 2
(C) 4 (D) 0.5

SECTION B: ASSERTION - REASON QUESTIONS (Q.16 TO Q.20)

[5 MARKS]

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.16 Assertion (A): When a student holds a heavy 20 kg school bag motionless on his shoulder for 30 minutes, no scientific work is done.

Reason (R): Work is done only when a force produces a non-zero displacement in the object.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.17 Assertion (A): The total mechanical energy of a freely falling body under gravity remains constant at every point in its fall.

Reason (R): As the body falls, its potential energy decreases by the exact amount by which its kinetic energy increases.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.18 Assertion (A): A pair of tweezers or sugar tongs is a Class 3 lever with mechanical advantage less than 1.

Reason (R): In a Class 3 lever, the Effort is applied between the Fulcrum and the Load, acting as a speed/distance multiplier rather than a force multiplier.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.19 Assertion (A): Power is the scalar product of force and velocity.

Reason (R): Power measures the rate at which energy is consumed or work is executed.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.20 Assertion (A): A fast-moving bullet possesses high kinetic energy.

Reason (R): Kinetic energy of an object depends inversely on its velocity.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

SECTION C: CASE-BASED / EXPERIMENTAL QUESTIONS (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL SETUP:

Case Study: Hydroelectric Power Station Energy Transformations

At a hydroelectric dam, water stored in a huge high-altitude reservoir at a height of 50 m falls through massive pipes (penstocks) onto turbine blades located at the base of the dam. The gushing water spins the turbines at high speed. The spinning turbine drives an electric generator that produces electricity, which is then transmitted via high-voltage transmission lines to illuminate homes and power industries.

Q.21 What form of energy is stored in the reservoir water high above the dam?

- (A) Kinetic energy
 (B) Gravitational potential energy
 (C) Chemical energy
 (D) Thermal energy

Q.22 As water rushes down the penstock pipe towards the turbine, the energy transformation occurring is:

- (A) Kinetic energy into chemical energy
 (B) Potential energy into kinetic energy
 (C) Electrical energy into potential energy
 (D) Nuclear energy into thermal energy

Q.23 If 2000 kg of water falls onto the turbine blades every second from a height of 50 m ($g = 10 \text{ m/s}^2$), the potential energy converted per second (power) is:

- (A) 100 kW (B) 500 kW
(C) 1000 kW (1 MW) (D) 2000 kW

Q.24 What device converts the kinetic energy of the rotating turbine shaft into electrical energy?

- (A) Electric motor (B) Electric generator / Alternator
(C) Step-down transformer (D) Electric battery

Q.25 Which fundamental universal law confirms that no new energy is created, only transformed from one form to another during this process?

- (A) Newton's first law of motion (B) Law of conservation of energy
(C) Kepler's third law (D) Hooke's law of elasticity

CBSE CANDIDATE OMR ANSWER RESPONSE GRID

01	(A) (B) (C) (D)	02	(A) (B) (C) (D)	03	(A) (B) (C) (D)	04	(A) (B) (C) (D)	05	(A) (B) (C) (D)
06	(A) (B) (C) (D)	07	(A) (B) (C) (D)	08	(A) (B) (C) (D)	09	(A) (B) (C) (D)	10	(A) (B) (C) (D)
11	(A) (B) (C) (D)	12	(A) (B) (C) (D)	13	(A) (B) (C) (D)	14	(A) (B) (C) (D)	15	(A) (B) (C) (D)
16	(A) (B) (C) (D)	17	(A) (B) (C) (D)	18	(A) (B) (C) (D)	19	(A) (B) (C) (D)	20	(A) (B) (C) (D)
21	(A) (B) (C) (D)	22	(A) (B) (C) (D)	23	(A) (B) (C) (D)	24	(A) (B) (C) (D)	25	(A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & DETAILED EXPLANATIONS

Q. No.	Correct	Conceptual Explanation & Syllabus Rationale
Q.1	(B)	$\$W = mgh = 15 \times 10 \times 1.5 = 225\$ \text{ J}.$
Q.2	(B)	Gravitational force is directed towards the Earth's center (perpendicular to tangential displacement, $\theta = 90^\circ$). Since $\$W = F s \cos 90^\circ = 0\$$, work done is zero.
Q.3	(B)	$\$E_k = \frac{1}{2}mv^2\$$. Since $\$E_k \propto v^2\$$, doubling speed ($2v$) makes kinetic energy $\$(2)^2 = 4\$$ times.
Q.4	(B)	Friction acts in the direction opposite to displacement ($\theta = 180^\circ$). Therefore, $\$W = F s \cos 180^\circ = -F s\$$ (negative work).
Q.5	(A)	$\text{Power} = \frac{\text{Energy}}{\text{Time}} = \frac{1000 \text{ J}}{10 \text{ s}} = 100 \text{ Watts}.$
Q.6	(B)	$1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ s} = 3.6 \times 10^6 \text{ J}.$
Q.7	(B)	By conservation of mechanical energy, final $\$E_k = \text{initial } E_p = mgh = 2 \times 10 \times 20 = 400\$ \text{ J}.$
Q.8	(B)	Mechanical Advantage $\$= \frac{\text{Load}}{\text{Effort}}\$$. A machine with $\$MA > 1\$$ acts as a force multiplier.
Q.9	(A)	Class 1 lever has the Fulcrum in the middle (e.g., seesaw, crowbar, scissors). Class 2 has Load in the middle; Class 3 has Effort in the middle.
Q.10	(B)	In a wheelbarrow and nutcracker, the Load is positioned between the Fulcrum and the Effort (Class 2 lever, $\$MA > 1\$$).
Q.11	(A)	$\$W = mgh = 200 \times 10 \times 30 = 60,000\$ \text{ J}.$ Time $\$t = 2 \times 60 = 120 \text{ s}.$ $\$P = \frac{60000}{120} = 500\$ \text{ W}.$
Q.12	(B)	Work done against restoring spring force is stored as elastic potential energy due to configuration change.
Q.13	(B)	$\$W = F s \cos 60^\circ = 10 \times 5 \times 0.5 = 25\$ \text{ J}.$
Q.14	(B)	A chemical battery/cell converts stored chemical energy into electrical energy through redox reactions.
Q.15	(A)	A single fixed pulley does not multiply force ($\$MA = 1\$$); it simply changes the direction of the applied effort to make lifting convenient.
Q.16	(A)	Because displacement $\$s = 0\$$, scientific work $\$W = F \times s = 0\$$, despite physiological muscular fatigue.
Q.17	(A)	In the absence of air resistance, mechanical energy $\$E_k + E_p\$$ is conserved; gravitational potential energy converts completely into kinetic energy.
Q.18	(A)	In Class 3 levers, effort arm is shorter than load arm ($\$MA < 1\$$), requiring more effort force but providing precision and amplified displacement.
Q.19	(B)	Both are true ($\$P = W/t = F \cdot v\$$), but the second statement is the definition rather than the derived proof of $\$F \cdot v\$$.
Q.20	(C)	Assertion is true. Reason is false because kinetic energy is directly proportional to the square of velocity ($\$E_k = \frac{1}{2}mv^2\$$).
Q.21	(B)	Water elevated to height $\$h\$$ possesses gravitational potential energy given by $\$mgh\$$.
Q.22	(B)	As water loses height, its potential energy is converted into kinetic energy of flowing water.
Q.23	(C)	$\text{Energy per second} = mgh = 2000 \times 10 \times 50 = 1,000,000 \text{ J/s} = 1000 \text{ kW} = 1 \text{ Megawatt}.$
Q.24	(B)	A generator uses electromagnetic induction to convert rotational kinetic energy into electricity.
Q.25	(B)	The law of conservation of energy states that energy cannot be created or destroyed, only transformed.